# From data to a decision

A stats primer: mean, standard deviation, standard error and confidence intervals for one and two groups, then testing a null hypothesis

Every experiment ends with numbers and a question: *is there really a difference, or did we get lucky?* This primer builds the answer one step at a time. Each step is a small idea, and each one uses the step before it. Read it in order; it takes about 20 minutes.

All the examples use made-up **reaction times** (how many milliseconds people take to press a key), produced by the **stats lab**, a small simulation in this folder. Because the simulation knows the true answer, you can check what a study finds against the truth, which you can never do with real data. Every number and figure below comes from a stats lab command you can run yourself:

**Running the stats lab**

```
python3 stats_lab/stats_lab.py sample
```

(on Windows, `py` instead of `python3`). Or, in Claude Code, opened in the task folder, ask: "Run the stats lab *sample* command and explain the output." Add `--seed` with the number given in each section to get exactly the numbers printed here; leave it out to run a new study.

## 1 · One group

A study measures 20 people. The simulation draws them from a population whose true mean is 300 ms and whose true standard deviation is 45 ms. The study, of course, does not know that. Here are its 20 reaction times, in ms:

```
216, 236, 259, 268, 273, 283, 284, 291, 292, 306, 316, 324, 326, 326, 328, 345, 346,
347, 350, 382
```

### The mean (M): where the data sit

Add up the values and divide by how many there are:

$$M = \Sigma x / n = 6{,}098 / 20 = \mathbf{304.9\ ms}$$

Σ ("sigma") means "add up", x is each value, and n is the number of people. The study's mean (304.9) is close to the true mean (300) but not equal to it. A different 20 people would give a slightly different mean. Keep that thought: it is what the rest of this primer is about.

### The standard deviation (SD): how spread out the people are

Take each person's distance from the mean, square it (so negative and positive distances do not cancel), average the squares, and take the square root to get back to milliseconds:

$$SD = \sqrt{[\ \Sigma(x - M)^2 / (n - 1)\ ]} = \mathbf{41.9\ ms}$$

The SD is roughly the typical distance of a person from the mean. Dividing by n − 1 instead of n makes up for the fact that a sample sits a little closer to its own mean than to the true one. For bell-shaped data, about two thirds of people fall within 1 SD of the mean (here 14 of 20) and about 95% within 2 SD.

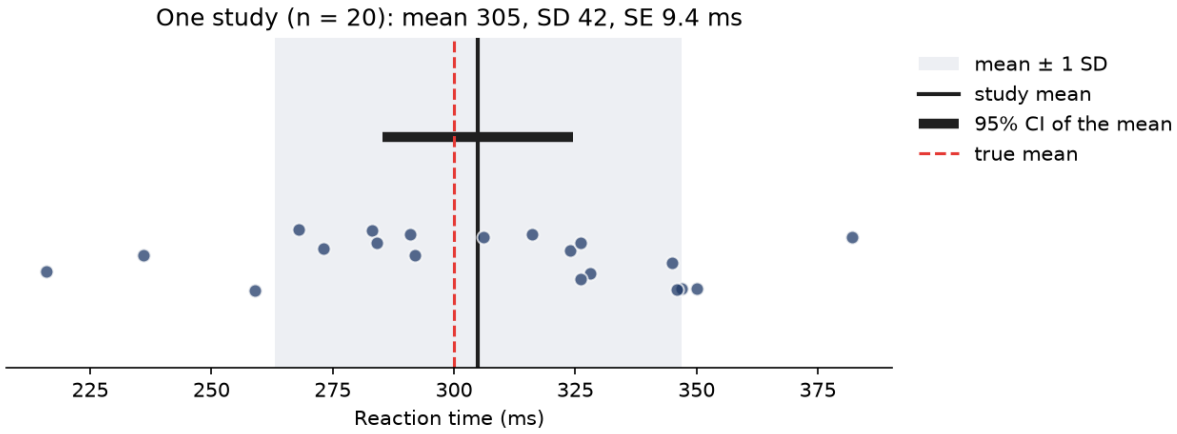



Figure 1. The 20 reaction times (dots), the study mean (black line), the band from 1 SD below to 1 SD above the mean (gray), the 95% confidence interval of the mean (black bar; see below) and the true mean (red), which only a simulation can show. `sample --seed 16`

## The standard error (SE): how precise the mean is

The SD describes *people*. To judge the *mean*, we need to know how much it would change if we ran the study again with different people. The stats lab can actually do that: it repeats the same study 10,000 times and collects the 10,000 means (Figure 2). Individual people spread out with an SD of about 45 ms; the means spread out with an SD of only about 10 ms. That spread of the means is the standard error. You do not need 10,000 studies to estimate it; one study and a formula will do:

$$SE = SD / \sqrt{n} = 41.9 / \sqrt{20} = \mathbf{9.4\ ms}$$

In the 10,000 simulated repeats, the means actually varied by 9.9 ms, and the formula with the true SD gives 45 / √20 = 10.1 ms. Because n sits under a square root, a study four times as large has half the standard error: more people, more precise mean, but with diminishing returns.

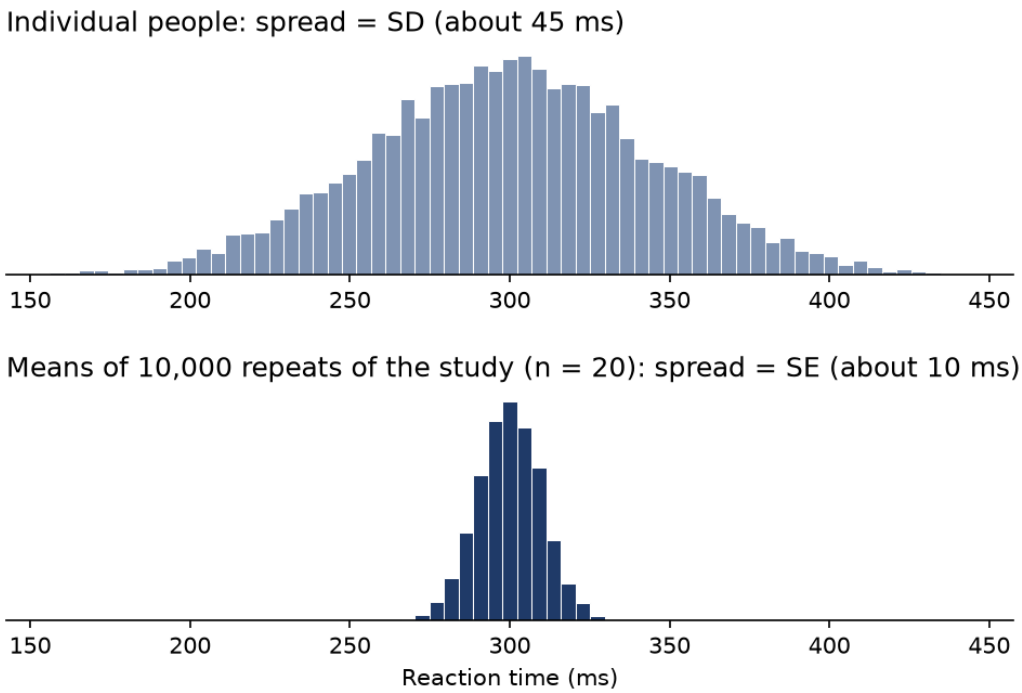



Figure 2. Top: individual reaction times. Bottom: the means of 10,000 imaginary repeats of the same 20-person study, on the same axis. `repeat --seed 11`

**Do not mix them up**

**SD** answers "how different are people from each other?" It does not shrink as you test more people.

**SE** answers "how far is my mean likely to be from the true mean?" It shrinks as you test more people. Error bars in papers can show either, so always check which one.

## The 95% confidence interval (CI): a range for the true mean

A mean on its own hides its uncertainty. A confidence interval puts a range around it, about two standard errors either side:

$$95\%\ \text{CI} = M \pm t^* \times SE = 304.9 \pm 2.09 \times 9.4 = \mathbf{[285.3, 324.5]\ ms}$$

t* ("t critical") is close to 2. It is a little larger for small samples, because their SD is itself uncertain. It depends on the degrees of freedom, df = n − 1 for one group:

| n | 5 | 10 | 20 | 50 | 100 | very large |
|---|---|---|---|---|---|---|
| t* for a 95% CI | 2.78 | 2.26 | 2.09 | 2.01 | 1.98 | 1.96 |

**What "95%" means.** If you ran the study many times and computed a CI each time, 95% of those intervals would contain the true mean. In 100 simulated studies, 95 did (Figure 3). Any one interval either contains the truth or misses it, and from inside a single study you cannot tell which. So the 95% describes the method, not one particular interval.

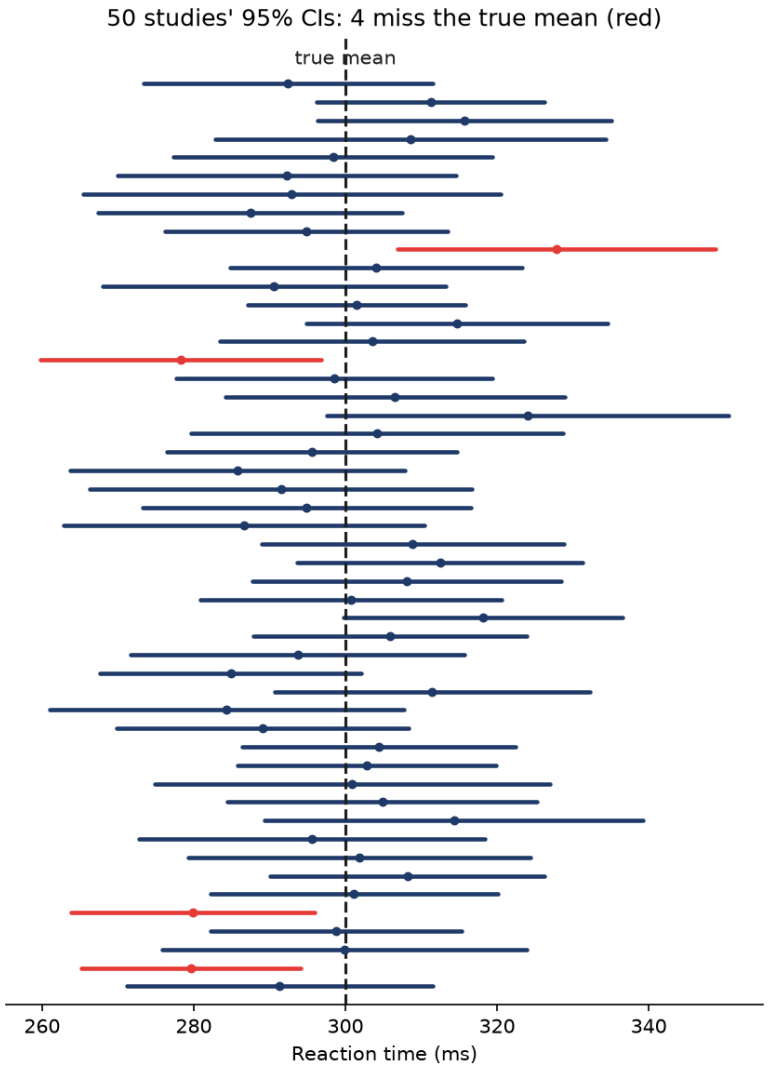



Figure 3. The 95% CIs of the first 50 of 100 simulated studies. Four miss the true mean (red). Across all 100, 95 contained it. `coverage --seed 5`

**Try it**

```
python3 stats_lab/stats_lab.py coverage --n 5
```

Small studies give wide intervals, large studies narrow ones, but about 95% contain the truth either way.

*In Claude Code:* "Run the stats lab coverage command with 5 people per study, then with 100. What changes, and what stays the same?"

# 2 · Two groups

Now a question: are people slower after a short night? Group A slept 8 hours and Group B slept 5 hours, 20 different people in each group. In the simulation, Group B really is slower: its true mean is 340 ms against 300 ms for Group A, a true difference of 40 ms, with the same true SD of 45 ms.

Describe each group exactly as in Part 1:

| | n | Mean | SD | SE | 95% CI of the mean |
|---|---|---|---|---|---|
| Group A (8 h) | 20 | 298.7 | 52.0 | 11.6 | [274.4, 323.0] |
| Group B (5 h) | 20 | 330.8 | 44.4 | 9.9 | [310.0, 351.5] |

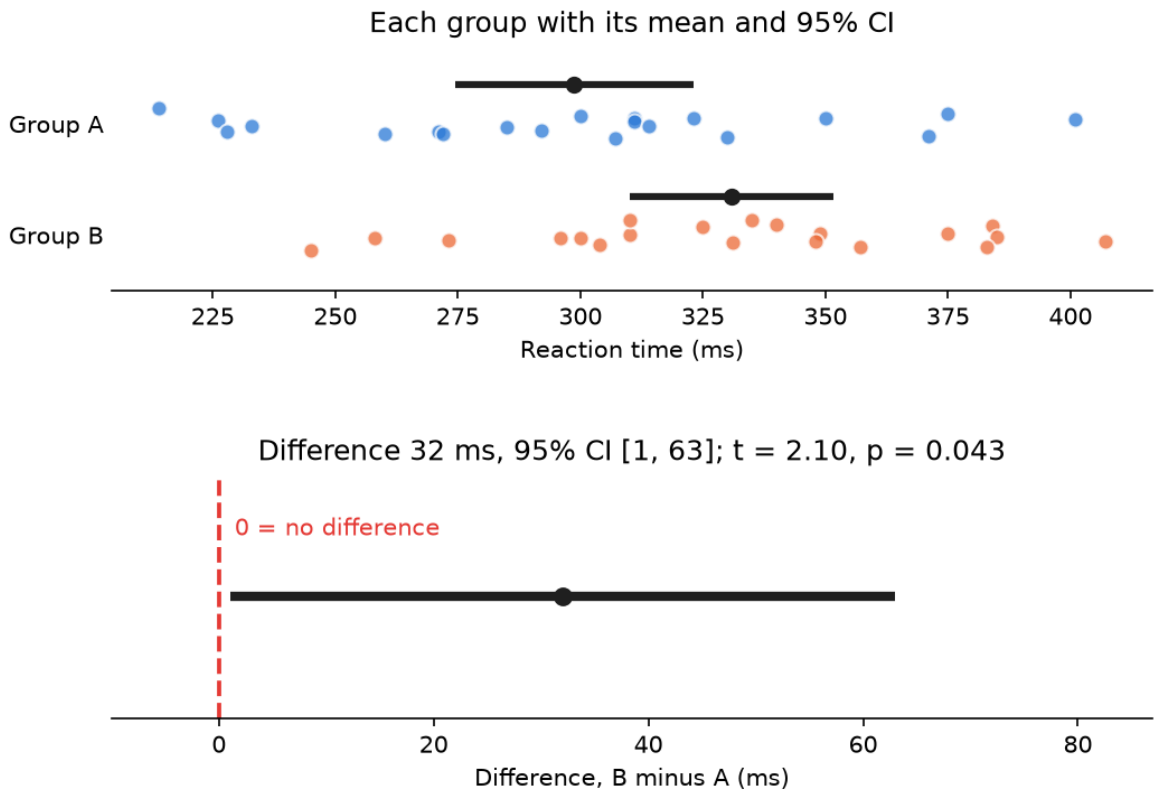



Figure 4. Top: each group's reaction times with its mean and 95% CI. Bottom: the difference between the means with its own 95% CI, next to 0 (no difference). `two --seed 4`

## The difference and its uncertainty

What we care about is the difference between the means: 330.8 − 298.7 = **32.1 ms**. Like any estimate it has a standard error. Both means are uncertain, so their uncertainties add up, like the sides of a right-angled triangle:

$$\text{SE of the difference} = \sqrt{(SE_A^2 + SE_B^2)} = \sqrt{(11.6^2 + 9.9^2)} = \mathbf{15.3\ ms}$$

and the 95% CI of the difference is built just like before: 32.1 ± 2.03 × 15.3 = **[1.1, 63.0] ms**. The interval is wide, but every value in it is above 0: the data point to Group B being slower, by somewhere between about 1 and 63 ms.

**A common trap**

The two groups' own CIs overlap (Group A goes up to 323.0, Group B starts at 310.0), yet the CI of the difference does not include 0. Overlapping group CIs do **not** mean "no difference". To judge a difference, look at the CI of the difference itself.

**Same people measured twice?**

Everything above assumes two separate groups of people. If the *same* people are measured twice (for example, before and after something), compute each person's difference first. Those differences form **one** group, so Part 1 applies: their mean, SD, SE and 95% CI, with df = n − 1. Measuring the same people twice usually gives a much smaller SE, because each person acts as their own comparison.

# 3 · Testing a null hypothesis

A confidence interval already tells you a lot. Hypothesis testing turns the same information into a yes-or-no decision, using a particular line of reasoning.

## The two hypotheses

**Null hypothesis, $H_0$:** there is no real difference; the true difference is 0. Any difference we see is chance, from which people happened to be in each group.

**Alternative hypothesis, $H_1$:** there is a real difference; the true difference is not 0 (in either direction).

We cannot prove $H_1$ directly. Instead we *assume $H_0$ is true* and ask: if it were, how surprising would data like ours be? If they would be very surprising, we reject $H_0$. It is like "innocent until proven guilty": $H_0$ is the presumption of innocence, and the data have to be convincing to overturn it.

## t: how many standard errors away from 0?

$$t = \text{difference} / \text{SE of the difference} = 32.1 / 15.3 = \mathbf{2.10}$$

A difference of 32 ms means little until you compare it with how much differences wobble by chance, which is exactly what the SE measures. Our difference is 2.10 standard errors away from 0. If $H_0$ were true, t would usually land between about −2 and +2 (Figure 5, top). Big t comes from a big difference, little spread between people, or many people.

## p: how surprising is our result if $H_0$ is true?

The **p-value** is the probability of getting a t at least as far from 0 as ours (in either direction) *if $H_0$ were true*. For t = 2.10 with df = 37.1, **p = .043**: if there were no real difference, only 4.3% of studies like this one would show a difference this large.

By convention we set a cut-off before looking at the data, **$\alpha$ = .05**. If $p < .05$ we reject $H_0$ and call the difference "statistically significant"; otherwise we do not reject $H_0$. Here p = .043, so we reject $H_0$. Notice that this matches the CI: **the 95% CI of the difference excludes 0 exactly when p < .05**. They are two views of the same calculation.

**What p is not**

- **Not** the probability that $H_0$ is true. It assumes $H_0$ is true.
- **Not** the size or importance of an effect. A tiny effect in a huge study can have a tiny p; the CI shows the size.
- p > .05 does **not** show there is no effect. It may mean the study was too small to detect it.

## Errors: false positives and power

The stats lab can run 1,000 studies where $H_0$ really is true (no difference at all). p fell below .05 in **44 of 1,000** (4.4%; Figure 5). Each of those is a **false positive**: α = .05 means accepting about a 5% risk of this whenever there is nothing to find. When there really is a 40 ms difference, p fell below .05 in **776 of 1,000** studies. That 78% is the study's **power**. The other 22% of studies missed a real effect, which is why p > .05 cannot show that an effect is absent.

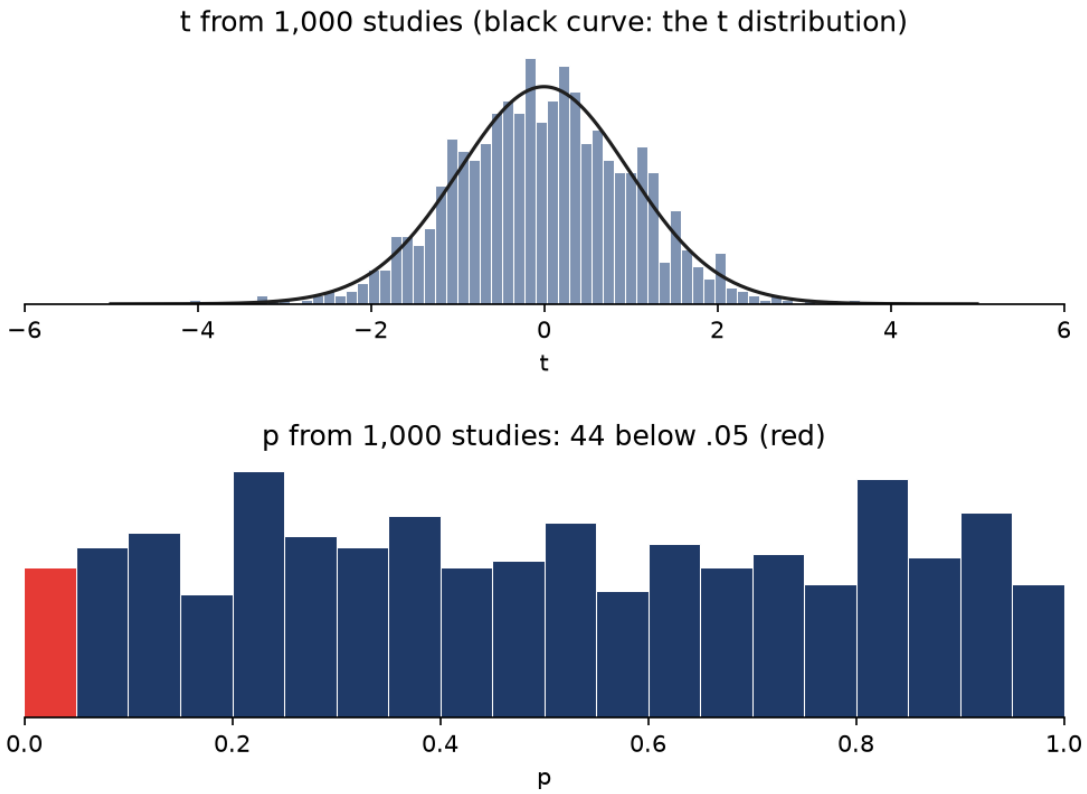



Figure 5. 1,000 two-group studies with no real difference. Top: their t values follow the t distribution (black curve). Bottom: their p values are spread evenly from 0 to 1; 44 fall below .05 (red). `many --seed 1`; power: `many --diff 40 --seed 2`

**Try it**

```
python3 stats_lab/stats_lab.py many --diff 20 --n 20
```

*In Claude Code:* "Use the stats lab to find out how many people per group I would need to detect a 20 ms difference 80% of the time."

# Cheat sheet

| Quantity | Formula | Answers the question | Example |
|---|---|---|---|
| Mean, M | $\Sigma x / n$ | Where do the data sit? | 304.9 ms |
| Standard deviation, SD | $\sqrt{[\Sigma(x - M)^2 / (n - 1)]}$ | How different are people from each other? | 41.9 ms |
| Standard error, SE | $SD / \sqrt{n}$ | How precise is the mean? | 9.4 ms |
| 95% CI of a mean | $M \pm t^* \times SE$ ($t^* \approx 2$) | Which true means are plausible? | [285.3, 324.5] |
| Difference of two means | $M_B - M_A$ | How big is the effect? | 32.1 ms |
| SE of the difference | $\sqrt{(SE_A^2 + SE_B^2)}$ | How precise is the difference? | 15.3 ms |
| 95% CI of the difference | $\text{difference} \pm t^* \times SE_{diff}$ | Which true differences are plausible? | [1.1, 63.0] |
| t | $\text{difference} / SE_{diff}$ | How many SEs is the difference from 0? | 2.10 |
| p | from t and df | If $H_0$ were true, how often would we see a t this extreme? | .043 |
| α | chosen in advance, usually .05 | How much false-positive risk will we accept? | .05 |

## Check yourself

1. A study doubles its sample from 25 to 100 people. What happens, roughly, to the SD? To the SE? To the width of the 95% CI?

2. Two groups' 95% CIs overlap a little. Can the difference still be significant? How would you check?

3. A study reports p = .30. A friend concludes the treatment "has no effect". What is wrong with that?

4. If 40 labs each test a treatment that truly does nothing, about how many will get $p < .05$?

5. Which tells you more about how big an effect is: p or the 95% CI of the difference? Why?

Answers: 1. SD stays about the same; SE halves ($\sqrt{4} = 2$); the CI gets about half as wide. 2. Yes; compute the CI of the difference (or the t-test). 3. $p > .05$ means the data do not rule out "no effect", not that there is none; the study may lack power, and the CI shows which effects are still plausible. 4. About 2 (5% of 40). 5. The CI: it is in the units of the measurement and shows the plausible range of the effect.